

Dataset Analysis and Choosing a Dataset

Part A (Store policy + Customer help desk):

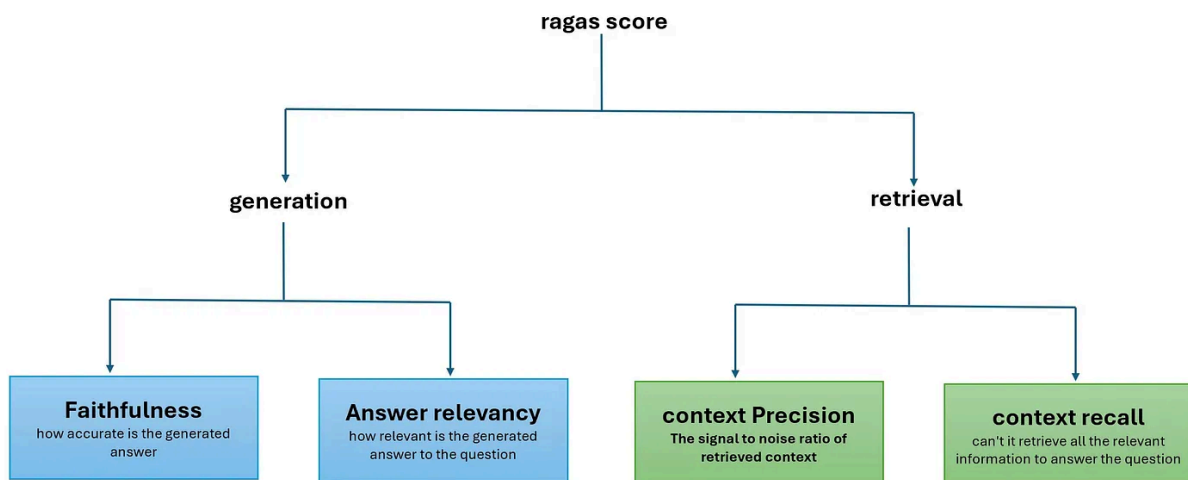
About the dataset:

A comprehensive policy framework was developed using ChatGPT, drawing references from established policy documents of leading clothing retailers such as H&M, Marks & Spencer (M&S), and Levi's. The formulated document encompasses key operational and customer-centric policies, including exchange, refund, and return procedures, garment recycling initiatives, loyalty and reward programs, as well as gifting policies.

Scoring Metric:

Retrieval-Augmented Generation Assessment Suite (RAGAS)

RAGAS is a specialized evaluation framework for RAG systems. It goes beyond basic overlap metrics (BLEU, ROUGE) by focusing on how well the generated answer is grounded in retrieved evidence and how relevant / faithful it is.



The main components are:

Faithfulness: Measures how factually correct the generated response is, relative to the information in the retrieved documents.

$$\text{Faithfulness} = \frac{\text{Number of Correct Facts in the Response}}{\text{Total Number of Facts in the Response}}$$

Answer Relevancy: Assesses how well the answer addresses the original query (i.e. relevance of concepts used).

$$\text{Answer Relevancy} = \frac{\text{Number of Relevant Concepts in the Response}}{\text{Total Number of Concepts in the Response}}$$

Context Precision: Among the retrieved passages, how many are actually relevant to the query (i.e. proportion of useful retrieval).

$$\text{Context Precision} = \frac{\text{Number of Relevant Sentences Retrieved}}{\text{Total Number of Sentences Retrieved}}$$

Context Recall: Of all the potentially relevant sections, how many relevant ones were successfully retrieved.

$$\text{Context Recall} = \frac{\text{Number of Relevant Sentences Retrieved}}{\text{Total Number of Relevant Sentences Available}}$$

Composite Score:

$$\text{RAGAS} = \frac{F + R + P_{\text{ctx}} + R_{\text{ctx}}}{4}$$

This composite score offers a balanced measurement of how well our system retrieves supporting evidence and generates a relevant, factual answer in the domain of policy / FAQ documents. We will also analyze component scores individually to identify weaknesses (e.g. low context recall might indicate retrieval gaps, while low faithfulness might indicate hallucination in generation).

Dataset:

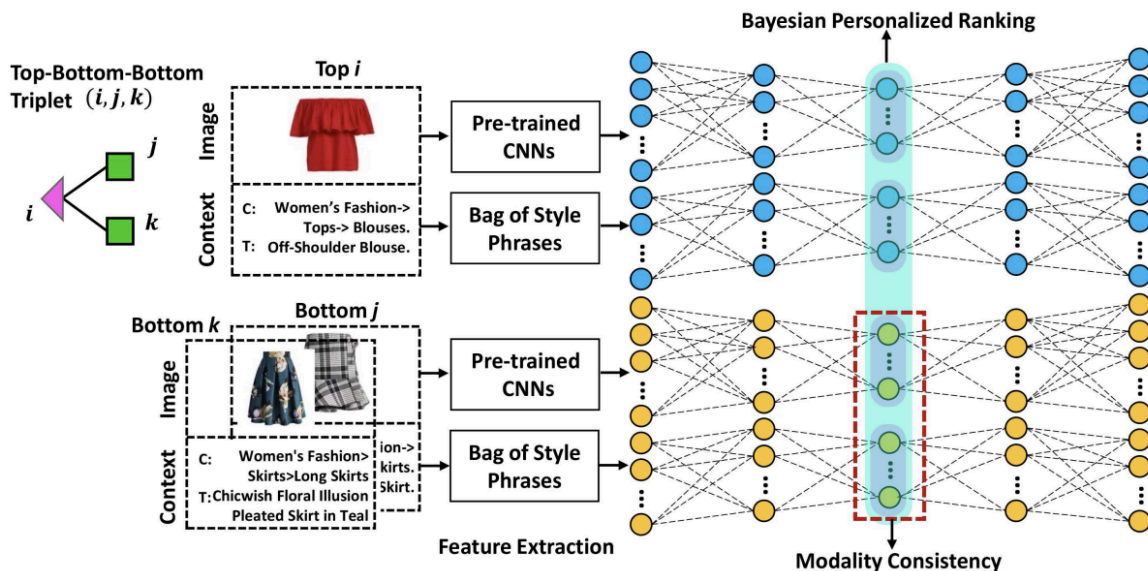
check document 'faq_and_policy.pdf'

link: https://github.com/maddy-git/group-5-DS-AI-Lab-Project/blob/main/milestones/milestone%202/faq_and_policy.pdf

Part B (AI stylist):

Literature review:

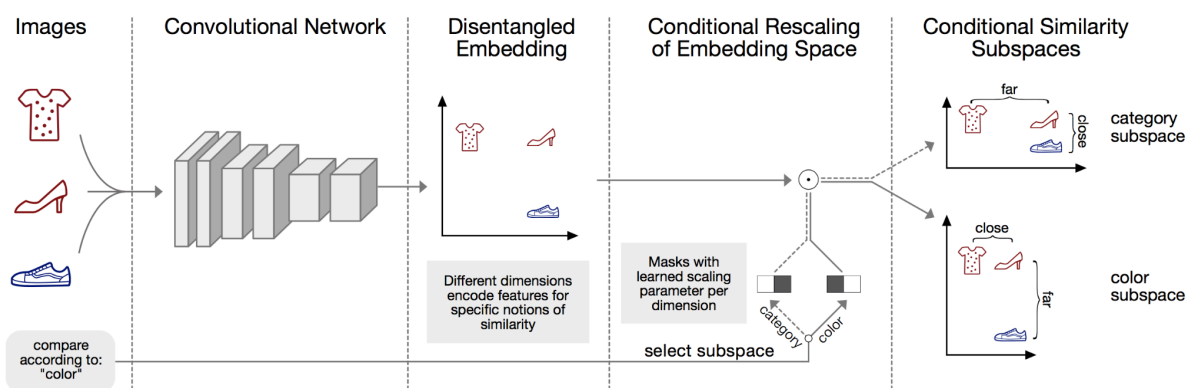
1. NeuroStylist (<https://liqiangnie.github.io/paper/p753-song.pdf>)
 - a. Used image as well as bag of style text input
 - b. Trained to cluster matching top and bottom closer.
 - c. Used triplet (input + pair + non-pair) items for calculating loss
 - d. Used AUC and MRR metrics for performance



2. Conditional Similarity Networks

(<https://github.com/andreasveit/conditional-similarity-networks?tab=readme-ov-file>)

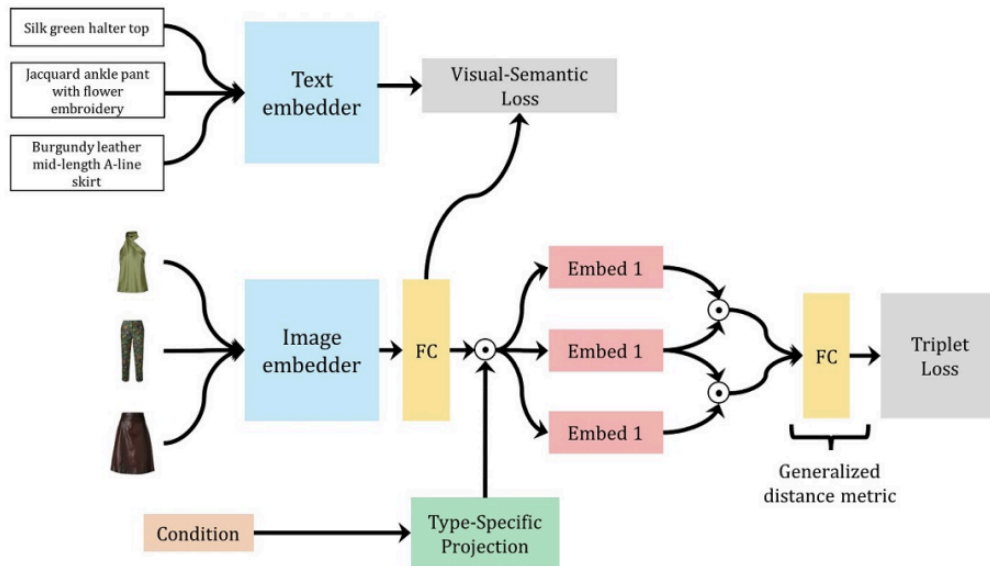
- a. Normally embedding the products clusters similar categories together.
- b. This research paper focuses on clustering compatible products along a specific dimension



3. Learning Type-Aware Embeddings for Fashion Compatibility

(<https://arxiv.org/pdf/1803.09196>)

- Used both image and text for training the model
- Again triplets are used to make the model learn about compatibility



Inference from the literature reviews:

- All the papers we researched focused on suggesting a good pair given a top/bottom and some models also took into account user preference. But no model tried to take into account user defined filters during search.
- With respect to the dataset part, for filtering we require lots of features on which filters can be applied. The datasets which have features with it do not provide any compatible pairs data. If compatible pairs data is present, then features are not available.
- So, we are going to use the iMaterilist dataset, which has features with it. With respect to clustering compatible products using images, we won't be able to perform due to unavailability of compatible pairs data. Instead we will try to use text information alone and will build a suggestion system.

So, our suggestion system is going to use text features alone and will suggest clothes based on user inputs.

Scoring Metric (Tentative):

1. In the research papers, models trained would find a compatible top/bottom, but filters were not applied. But in our project filtering on user query is going to be a major part and so we will define a custom scoring metric.
2. Tentative scoring metric (for illustration)

$$\text{score} = 0.2 * \text{color} + 0.2 * \text{material} + 0.2 * \text{gender} + 0.2 * \text{category} + 0.2 * \text{compatibility}$$

3. Any filter that is not provided in the user query will be ignored during score calculation and we will scale the score based on the filters available.
4. For compatibility we experimented with metrics like color harmony, but if we get access to a trained model from the research sources we have referred to we may use that.

Model Performance Test (for Baseline and Benchmarks):

1. Initially, we will not give the model much context about the nature of the data and specific instructions which it needs to follow for better search. We will use this as a baseline.
2. Subsequently we will define steps to follow for better search and also will try to increase the llm's understanding of fashion (compatibility text based) and will compare the performance with the baseline.
3. Additionally we will try different embedders and compare the performance.

Dataset Source:

Dataset Used: iMaterialist Fashion Attribute Dataset (CVPR 2018, FGVC5 Challenge)

License/Usage Terms: The dataset is publicly available and may be used for academic research and educational purposes under the competition's data usage rules. Redistribution or commercial use of the data is not permitted.

(<https://www.kaggle.com/competitions/imaterialist-challenge-fashion-2018/rules>)

Citation:

Guo, Sheng, Weilin Huang, Xiao Zhang, Prasanna Srikhanta, Yin Cui, Yuan Li, Matthew R. Scott, Hartwig Adam, and Serge Belongie. "The iMaterialist Fashion Attribute Dataset." arXiv preprint arXiv:1906.05750 (2019).

Brief about data:

We are utilizing the iMaterialist dataset available on Hugging Face ([Marqo/iMaterialist](#)) as the foundation for our product catalog analysis.

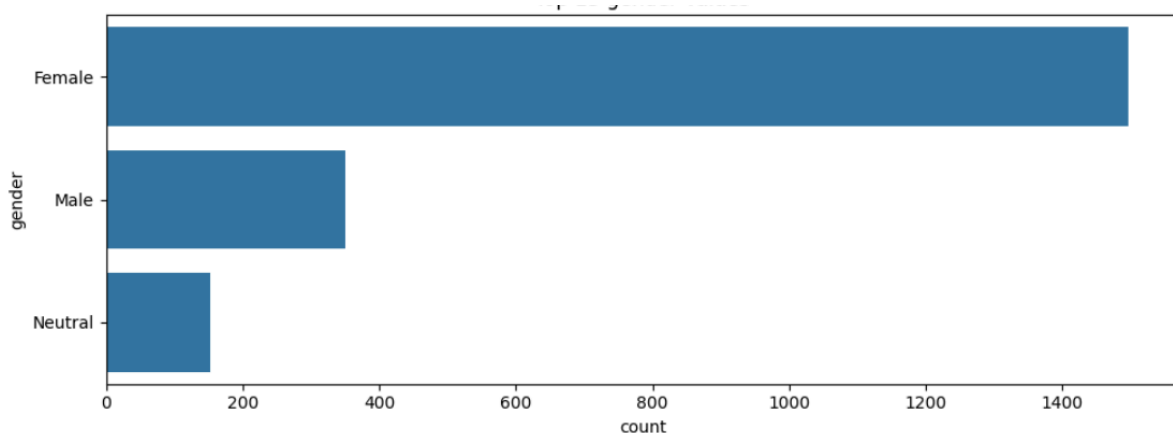
The dataset provides a rich collection of fashion-related attributes, making it ideal for understanding product diversity, and visual characteristics within the apparel domain.

The iMaterialist dataset is very large(70gb+ and 721k+rows) so we have only used an extract of first 2000 rows present in it to perform this EDA

EDA on iMaterialist

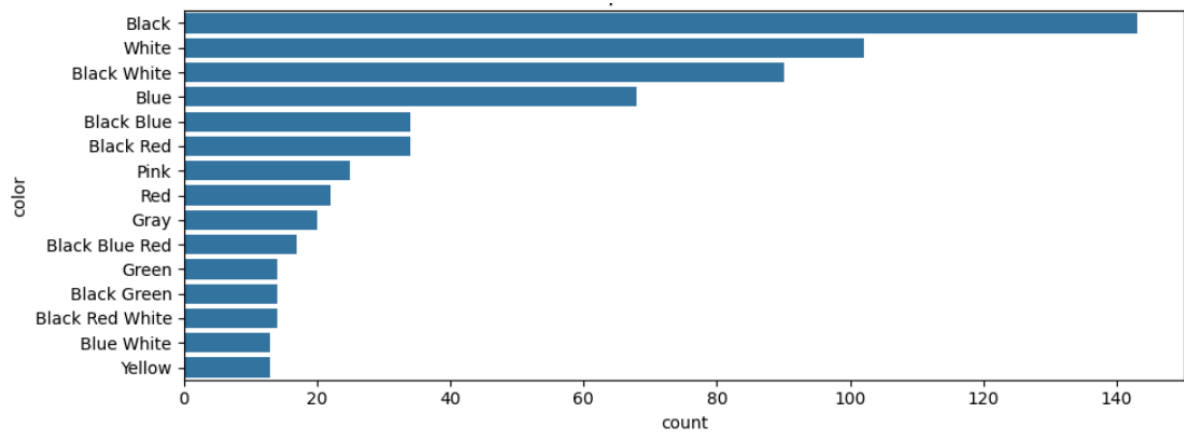
Link:<https://github.com/maddy-git/group-5-DS-AI-Lab-Project/blob/main/milestones/milestone%202/EDA.ipynb>

1. Gender Balance



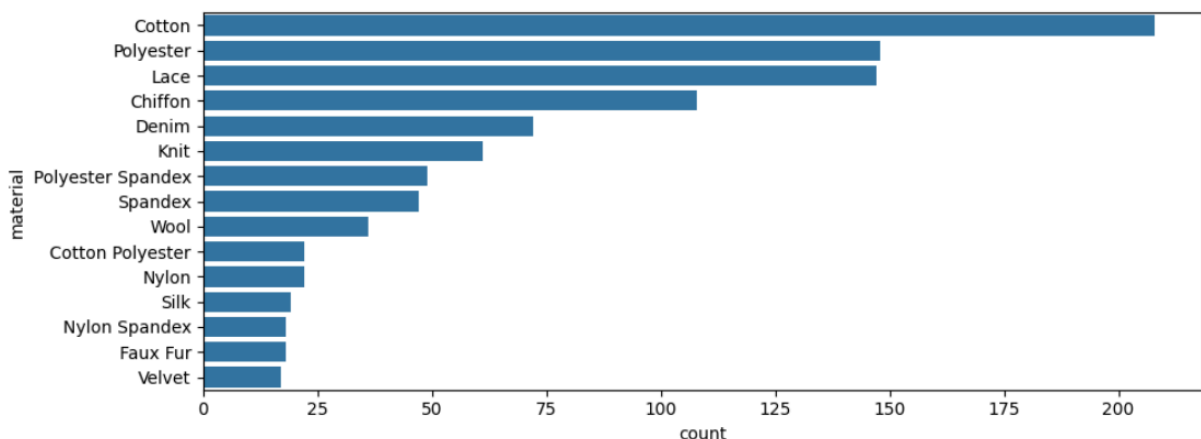
1. Female-oriented products dominate the dataset, while male and unisex categories account for a significantly smaller share.
2. The strong representation of female fashion items suggests that the dataset is primarily tailored toward women's apparel, which aligns with global retail trends — women's clothing consistently makes up the majority of fashion inventories and online product listings.
3. This gender imbalance is typical in fashion datasets, as women's categories tend to have greater product variety in terms of color, pattern, style, and seasonal trends.
4. The balanced presence of male and neutral items, however, ensures that the dataset still provides useful diversity for cross-gender product analysis and recommendation tasks.

2. Color distribution:



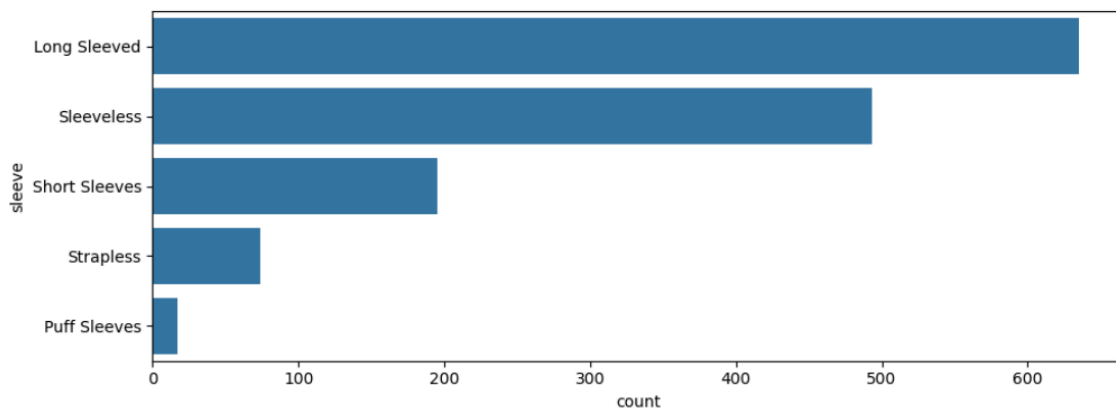
1. The most frequent colors are Black, White, and combinations like Black & White or Black & Blue.
2. Dark neutral tones such as black and white dominate because they are versatile and timeless, making them highly popular in both casual and formal wear.
3. This aligns with global fashion trends — black is considered universally flattering, and brands often offer multiple monochrome variations of the same design to appeal to wider audiences.

3. Material distribution:



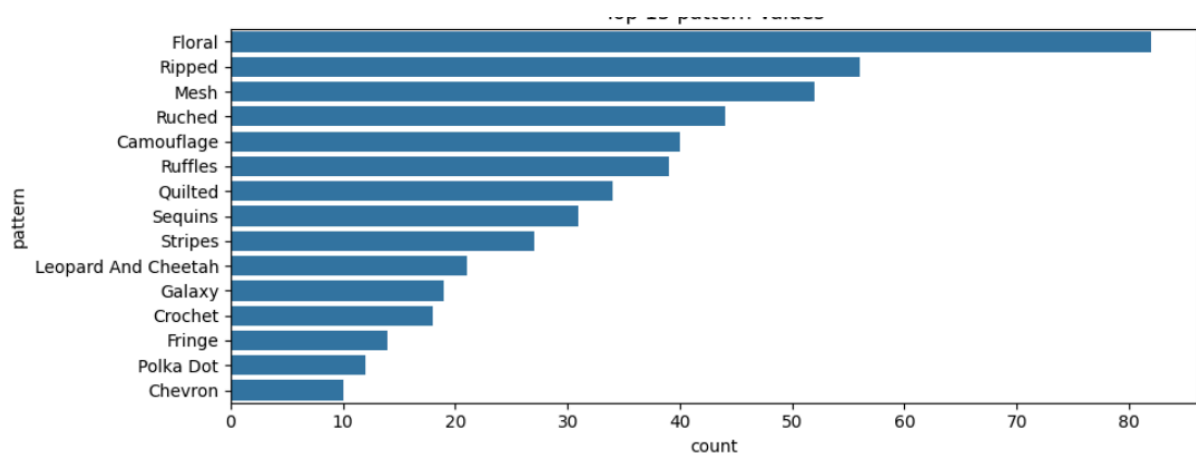
1. The most common materials are Cotton and Polyester, followed by Lace, Chiffon, and Denim.
2. Cotton and polyester are preferred due to their comfort, affordability, and suitability for mass production.
3. Lace and chiffon indicate a strong representation of feminine and dressy garments, suggesting that the dataset likely includes a high number of women's apparel items. The mix of natural (cotton) and synthetic (polyester, spandex) fibers also shows balance between comfort and elasticity-based performance materials.

4. Sleeve Type distribution:



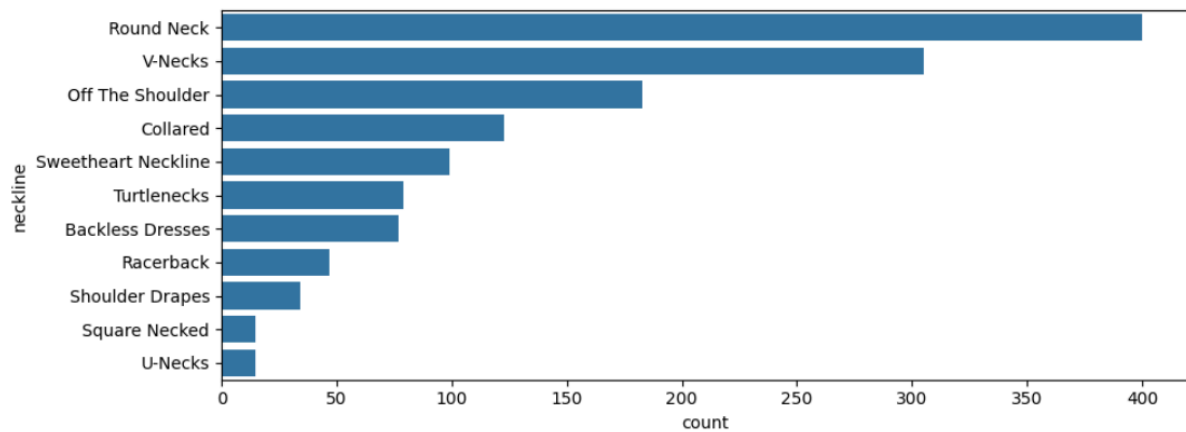
Most items have long sleeves, followed by sleeveless and short-sleeved designs.

5. Pattern distribution:



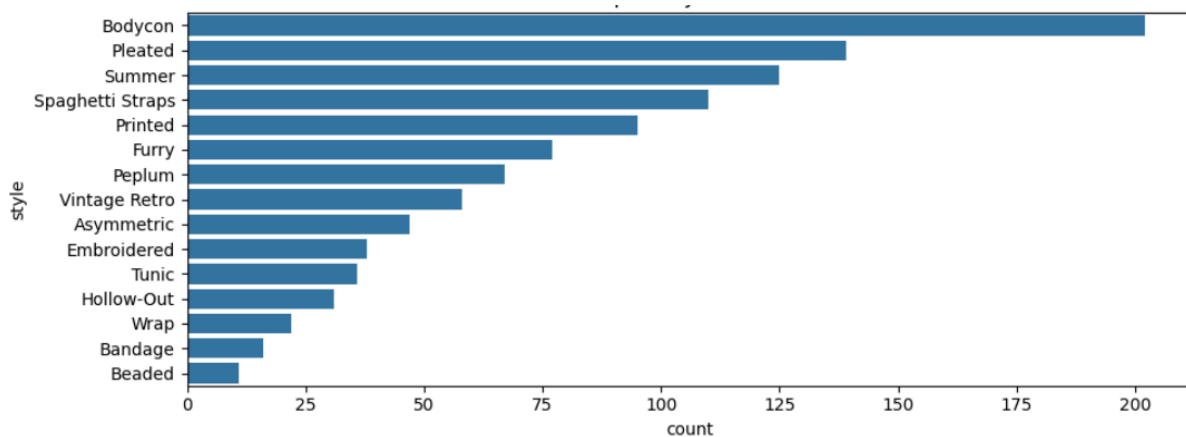
1. The most frequent patterns are Floral, Ripped, Mesh, and Ruched designs. Floral patterns dominate as they are highly popular in women's fashion, particularly for dresses and blouses. Ripped and ruched patterns indicate the inclusion of trendy and youth-oriented styles like ripped jeans or ruched tops. This suggests the dataset captures a wide variety of modern and classic patterns, reflecting fast fashion diversity.

6. Neckline distribution:



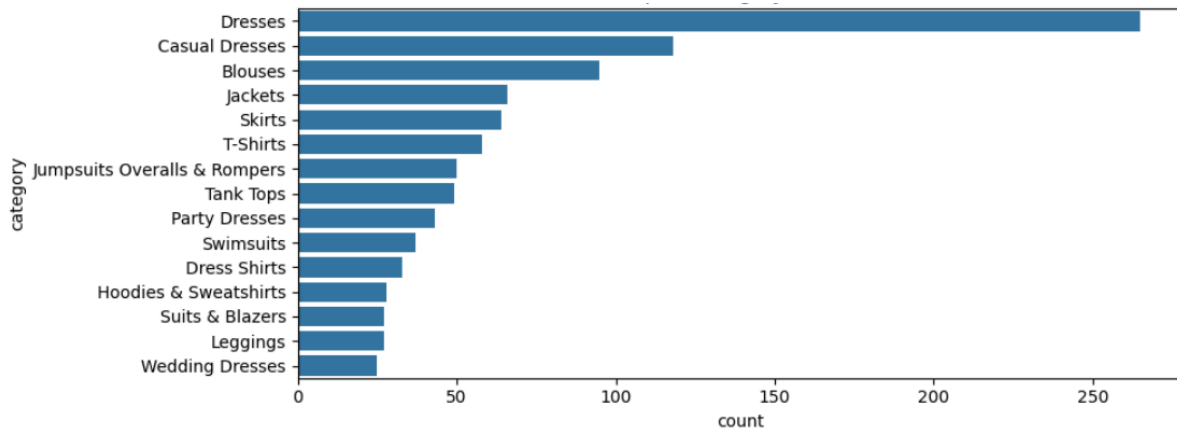
1. The leading neckline types are Round Neck and V-Neck, followed by Off-the-Shoulder and Collared styles.
2. Round and V-necks are the most common because they are simple, versatile, and work well across genders and occasions.
3. Off-the-shoulder and sweetheart necklines show a skew toward feminine designs, again consistent with fashion data where women's wear dominates

7. Style distribution:



1. The dataset shows that Bodycon, Pleated, and Summer styles are the most prevalent, followed by Spaghetti Straps, Printed, and Furry designs. Less frequent styles include Wrap, Bandage, and Beaded, which represent niche or specialized fashion categories.
2. The dominance of Bodycon and Pleated styles suggests a strong emphasis on modern and form-fitting women's fashion, commonly seen in urban and western apparel markets.
3. The high frequency of Summer and Spaghetti Strap styles reflects the popularity of seasonal, warm-weather clothing, which aligns with the overall gender distribution (female-dominated dataset).

8. Category distribution:



1. The highest number of items fall under Dresses, followed by Casual Dresses, Blouses, and Jackets.
2. The dominance of *Dresses* and *Casual Dresses* indicates that the dataset is heavily focused on women's apparel, which aligns with real-world fashion trends — women's categories typically occupy the largest share in online clothing catalogs.
3. This suggests that the dataset primarily represents female-oriented fashion segments such as everyday wear, office wear, and occasion dresses.
4. The presence of diverse categories like *Jackets*, *T-shirts*, and *Hoodies* also implies that it covers multi-seasonal clothing, suitable for both casual and semi-formal settings.